

Engineering Tripos Part IIB

Project Planning Form 2025/26

Students should submit this form to Group Administrators by **Friday 13 June 2025**. If plans evolve over the summer, please update and resubmit.

Name of student: Kaiyuan Bao

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College: Homerton

Director of Studies: Dr Timoleon Kipouros

Project is: Type (a)

Name of supervisor: Dr Jossy Sayir

Short title of project: MIMO techniques in 5/6G

Project reference code (e.g. B-ANO22-1 / B-ANO22-type(b)): F-js851-3*

Aims of project:

The aim of this project is to analyze Multiple-Input Multiple-Output (MIMO) techniques in 5G/6G systems. It will explore key concepts such as channel modeling, estimation, and coding, and will include simulations to support potential improvements in these techniques. The project is conducted under the co-supervision of CUED and Nokia Bell Labs, and may also provide insights into future technical specifications, potentially contributing to standardization efforts by organizations such as 3GPP.

Preparatory work for the long vacation:

Read selected chapters from textbooks and published papers recommended by the supervisors. Gain familiarity with the MATLAB 5G Toolbox. If possible, determine a specific research direction based on both personal interest and guidance from the supervisors.

Workshop/laboratory facilities needed:

No laboratory facilities are required for this project, as a personal laptop will be sufficient for all computational tasks.

Any special safety implications:

N/A

Signature of student: Kaiyuan Bao

Signature of supervisor:

